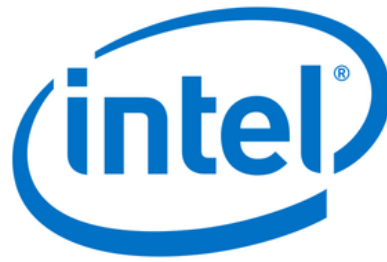


Item 1

Professional consultancy in BI and SM



1968-2006



2006-2020



2020-present

Student No.

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About Intel

Intel Corporation was founded in 1968 by Robert Noyce and Gordon Moore. It is known for its innovation in computer processing, particularly in the personal computer (PC) and server markets. Over the last few decades, Intel has experienced significant growth and expansion and has dominated the personal computing CPU market.

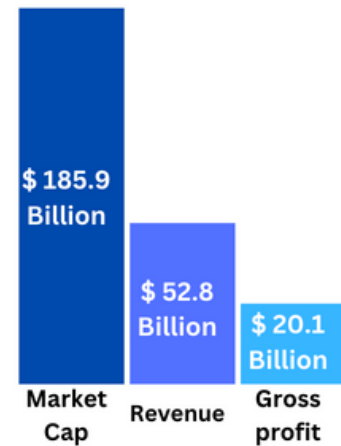


Fig 1: Business report (Intel, 2023b)



Fig 2: Multinational Corp (Intel, 2023b)

It has expanded to more than 10 countries and has over 131,900 employees (Intel annual report, 2021) which operate from offices from US, Europe, Middle East and South-East Asia. However, the rise of mobile computing, competition from AMD and ARM processors, have contributed to a maturing PC market and pressured Intel's traditional bread and butter.

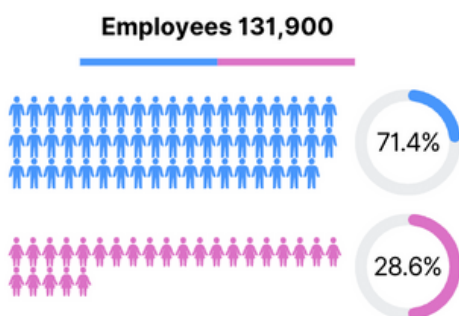


Fig 3: Diversity and inclusion (Intel annual report, 2021)

To maintain its dominance, Intel needs to navigate through these hurdles and explore new avenues. This is where consultancy expertise comes in, helping Intel identify future-proof opportunities, optimise its existing business, and adapt to the evolving tech landscape.

Why they need us

Intel is facing problems from multiple fronts which is reflected in their reduced revenue (-15% yoy), gross margin and growth from last several years (Ebrahim, 2021). The rise of smartphones and tablets has eroded Intel's traditional PC market share.

Production delays and manufacturing mishaps which happened with 7nm chip dealt a major blow to Intel (Gartenberg, 2021), leaving rivals like AMD and ARM to capitalise on the gap.

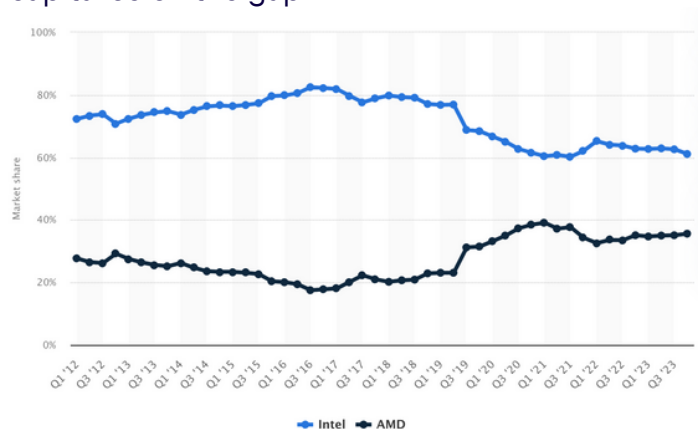


Fig 5: Losing market share to AMD (Intel, 2023)

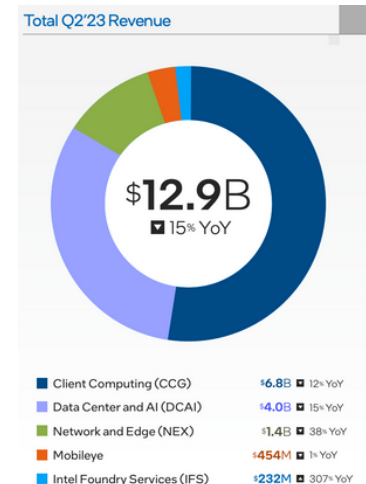


Fig 4: Q2'23 Revenue (Intel annual report, 2021)

Competition is intensifying from AMD's Ryzen processors and ARM's growing presence (Mujtaba, 2019) in servers applications is further pushing Intel on back foot, making them lose market share which is their primary cash cow.

Unprecedented growth of chip foundries like TSMC has limited buyers for Intel products.

As a prime example, one of their biggest chip buyer Apple has moved away from using Intel chips and started designing their own chips using third party vendors for fabrication.



Fig 6: Business report (Intel, 2023a)

About us

For this given project, we are acting as a consultancy firm that can provide strategic guidance and help Intel address high-level challenges, such as market disruptions, innovation, and overall corporate strategy.

As Intel faces complex technical and strategic challenges which require deep industry knowledge and specialised skills. We plan to thoroughly examine Intel's operations, create plans, and suggest practical actions to strengthen Intel's market position. We would like to follow expert consultancy model for this project. The expert consultancy model provides access to a highly qualified and experienced professionals with proven track records in areas such as chip technology, emerging markets, and business transformation.

Experts can quickly assess Intel's situation, identify key issues, and develop effective solutions leading to quicker decision-making and faster implementation, minimising potential disruptions and maximising ROI.

This approach provides Intel the flexibility to hire consultants whenever needed, ensuring that they have access to the right expertise when it's most important. The company can also scale back experts involvement as the project progresses or if their expertise is no longer required. This eliminates the upfront costs of a long-term contract, allowing Intel to save money and only pay for the expertise they actually use.

Overall, the expert consultancy model strikes a balance between flexibility, expertise, and customisation, making it a well-suited option for Intel's unique needs and challenges.

Product offerings



Fig 7: Products and services (intel.com, 2023)

Intel Corporation offers a wide range of products and services in the technology and semiconductor industry.

For consumers:

- CPUs and GPUs - It provides a range of CPUs for laptops, desktops, and gaming computers, giving users options to select their computing needs. Additionally, it offers integrated Intel graphics, that are found in limited laptops and desktops, providing graphical capabilities for everyday tasks and light gaming.
- SSDs and Wifi chipset - Apart from these Intel provides Wifi adapter chipset to provide network connectivity while Intel Optane which is a solid state drives that improve computer speed and responsiveness

For Businesses and Enterprises:

- Server and Data Center Solutions - Intel's Xeon processors and server technologies power data centres, cloud computing, and enterprise-level applications.
- IoTs and other solutions - It provides processors and edge computing solutions for embedded systems used to build smart devices. It also offers consulting and support services to assist businesses in optimising their IT infrastructure and technology investments.

Competitors

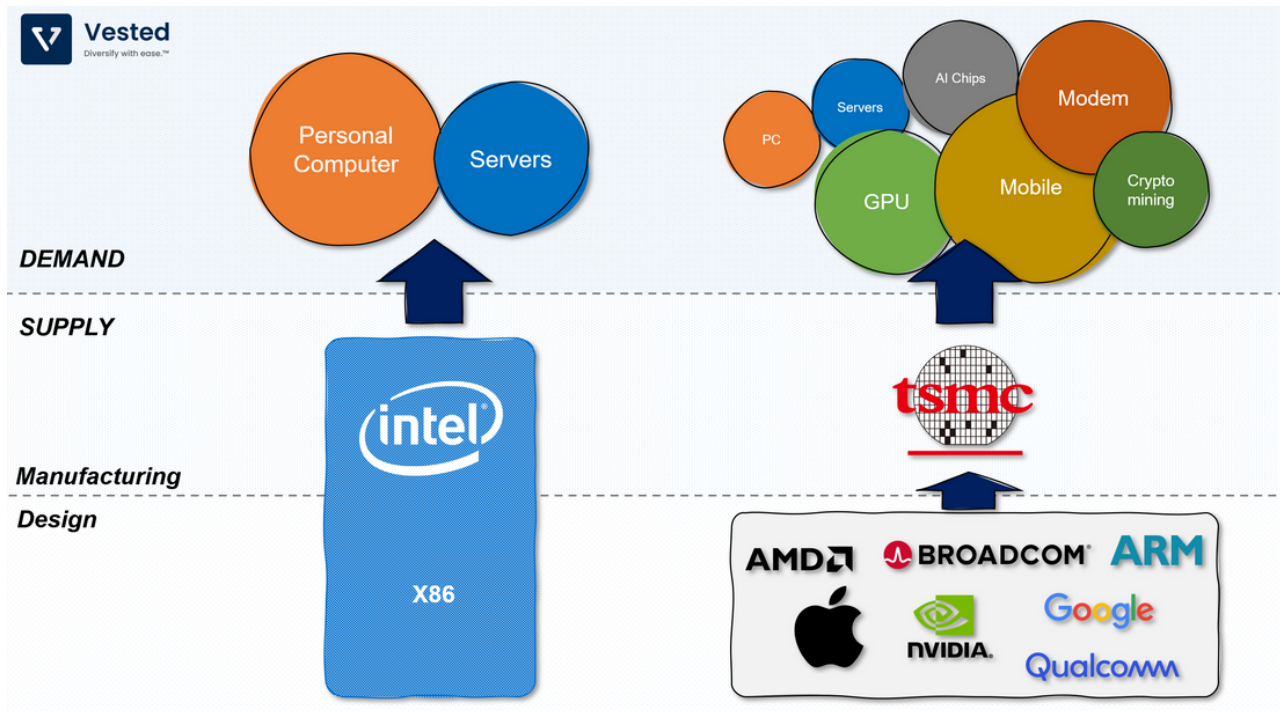


Fig 8: Products and services (Vested, 2022)

Intel primary faces competition on 2 fronts where its competitors are excelling

Chip design:

For decades, Intel led the charge in processor design, boasting the highest performance and efficiency. However, its recent stumbles in 7nm chip production have allowed AMD to steal the spotlight. AMD's Ryzen processors offer competitive performance at lower prices, making them a compelling choice for budget-conscious and performance-hungry users alike. Additionally, ARM-based processors are increasingly finding their way into servers and even laptops, further challenging Intel's dominance in the data center and mobile markets.

Chip Manufacturing:

Intel's manufacturing woes haven't helped their cause either. They've struggled with the transition to 7nm technology, resulting in delays and yield issues. Meanwhile, competitors like TSMC and Samsung have mastered 7nm and are already pushing towards 5nm and beyond, delivering smaller, faster, and more power-efficient chips. This manufacturing gap translates to higher costs for Intel, making it harder to compete on price and performance.

Competitors

Process timing		
Intel	Lead/Lag	Comp
32nm	++	28nm
22nm	++	20nm
14nm	+	16nm
10nm	--	7nm
Intel 4	-	5nm
Intel 3	-	TSMC 3nm
Intel 20A	-	TSMC 3nm

Fig 9: Manufacturing lag (Patel, 2023)

Competition is winning:

AMD: Ryzen processors offer excellent performance-per-dollar, winning over gamers and budget-conscious users. AMD's focus on chiplet design allows for faster development and adaptability to market needs.

TSMC has become world's leading chip foundry, and boasts superior manufacturing technology and efficiency, producing chips for various companies, including AMD and Apple, at competitive prices.

COMPANY	Q1 2022	Q2 2022	Q3 2022	Q4 2022
TSMC Revenue Growth* (YOY)	36%	37%	36%	27%
Intel Revenue Growth (YOY)	(1%)	(17%)	(15%)	(28%)

Fig 10: TSMC vs Intel (Alsop, 2022)

Underestimating the rise of smartphones and losing early advantage:

While Intel dominated the desktop and server chip market, its journey in mobile has been a story of missed opportunities and underestimation. This oversight has cost them dearly, allowing competitors like Apple and Qualcomm to establish themselves as leaders in the smartphone revolution. Many talented chip engineers left Intel for mobile-focused companies, further hindering their ability to compete in the space.

SWOT analysis

The SWOT analysis helps in identifying Intel's internal strengths and weaknesses, as well as external opportunities and threats. This provides balanced perspective which is crucial for understanding both the internal capabilities and external market conditions.

Strenghts

- Established market leadership in PC and server CPU markets.
- Strong financial resources for R&D and potential strategic moves.
- Valuable brand and intellectual property assets.

Weaknesses

- Failed entry into the mobile CPU market.
- Challenges in manufacturing advancements, losing lead to TSMC.
- Over-dependence on the shrinking PC market and volatile server market.

Opportunities

- Growth in AI, IoT, and cloud computing offers new markets for expansion.
- Potential collaborations or partnerships to improve manufacturing capabilities.
- Diversification into other semiconductor areas or new technology sectors.

Threats

- Intensifying competition from AMD and custom chips by cloud providers.
- Rapid technological advancements by competitors, especially in manufacturing.
- Geopolitical risks related to semiconductor manufacturing locations (limited to South East Asia)

3C's analysis

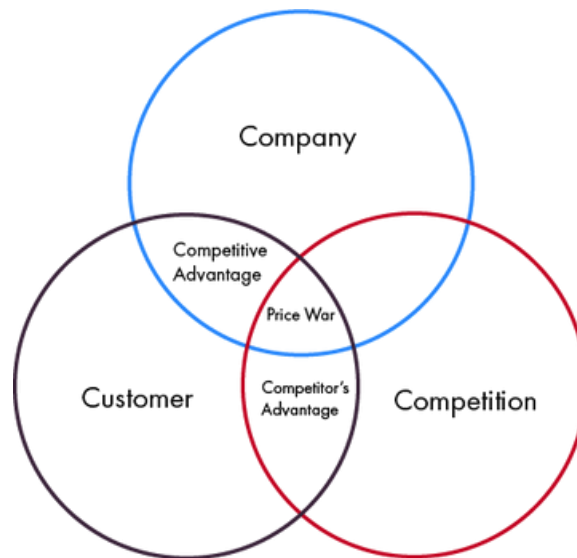


Fig 11: 3C's Company Customer and Competitor (Parametric, n.d.)

By analysing the company's capabilities, customer needs and preferences, and the competitive landscape, Intel can develop strategies that are customer-centric and competitively robust. The 3 C's model includes analysing intrinsic and extrinsic models to achieve a sustainable solution for the companies.

1. Company:

- Intel needs to navigate manufacturing setbacks and rethink its strategy, possibly shifting from being a design-centric to a manufacturing-centric company.
- Embrace diversification and new market opportunities to reduce over-reliance on traditional markets.

2. Customers:

- Evolving customer needs in mobile and cloud computing, with a shift towards energy efficiency and customisation.
- Cloud providers and data centre seeking more innovative and cost-effective solutions.

3. Competitors:

- AMD's rise in server CPUs and TSMC's manufacturing supremacy.
- New competitors in the chip design space, including companies like Apple and Nvidia.

BCG matrix



Fig 12: BCG matrix (Indeed, 2021)

The BCG (Boston Consulting Group) Matrix is a strategic tool used for portfolio analysis in business management. It helps companies like Intel analyze their various business units or product lines in terms of their relative market shares and growth rates. For Intel, this analysis is particularly useful to strategically allocate resources and make decisions about where to invest, develop, or divest.

Stars:

- Server CPUs: High market share but facing increasing competition and changing customer needs. This is a leading product in fast-growing markets. Star products generate strong revenues but also require continuous investment to maintain their position.

Question Marks:

- New ventures in AI, IoT, and autonomous systems: High growth potential but uncertain market position. This represents opportunities in growing markets, but where Intel doesn't have a strong position yet.

Cash Cows:

- PC CPUs: Dominant market share but in a declining market. These are mature products with a strong market presence. They are generating steady revenue with little need for investment.

Dogs:

- Mobile CPU market: Late and failed entry without significant presence. These are units with weak market presence in low-growth areas, often draining resources due to uncertainties

Strategic recommendation

The strategic recommendations to address the current challenges that it is facing can be divided into the following high level points:

Manufacturing Investment and Partnership:

- Intel currently lags behind TSMC in IC chip performance and size. It needs to follow a Lean innovation model to leap ahead its competition and this would require part of their team to operate in a startup mode.
- This team needs to be highly focussed and scope of autonomy and experimentation to revert with quick results.

Diversification into AI and IoT:

- As tech space is rapidly evolving, Intel's future growth depends on successful diversification into upcoming fields like AI and IoT. Though it is late to enter this segment, but it can set itself as a dominant player in emerging markets using its vast network of supply chain.

Strategic Collaborations:

- Intel holds IP in a number of key technologies, the is preventing it work with third party vendors as it can potentially lose its innovative edge in some areas. In such cases, the company can plan to make internal divisions that allow it collaborate with other firms where can supervise a clean handoff while working with competitors

Geopolitical Considerations and Domestic Manufacturing:

- After COVID, companies have realised that working in select location puts a major risk to logistics and can harshly affect the global supply chain. US govt. has already planned to provide subsidises (BusinessToday, 2023) for chip making companies to expand their manufacturing in US. Intel can jump on this opportunity and divert its expansion within the region.

Risks and KPIs

The associated risks associated with recommendations provided earlier can be categorised into following sections and their KPIs can be used to effectively track the progress.

- Budgeting large capital investment:
 - A shift in strategy for a manufacturing company is highly capital intensive. Though it might provide a better return in long run, but it would reduce the operating income for the coming quarters.
 - Entering into new markets will also require providing incentives to local sellers. A significant level of demand needs to be established before putting a large investment in developing economies.
- Diversification and unknowns
 - Entering into a new segment like IoTs or mobile chips will involve facing some unknowns of the sector which might add uncertainties and delay in process.

To effectively track the progress of strategic recommendations and whether they are inversely affected by the posed risks, we can follow the below KPIs and adjust its course as needed.

- Manufacturing Investment and Partnership:
 - Manufacturing cycle time (which is currently 2 years for next gen 5nm chip) will indicate progress in this area.
- Diversification into AI and IoT:
 - Success metrics for this, can be measured by the revenue growth in new segments and the number of successful product launches.
- Strategic Collaborations:
 - Number of partnerships and outputs like joint patents can serve as effective KPIs here.
- Geopolitical Considerations and Domestic Manufacturing:
 - Measuring percentage of domestically manufactured products vs foreign and the financial impact of reduced supply chain disruptions can be a key indicator here.

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THANK YOU!

Professional Consultancy in BI and SM - Group 1.8 (Item-2)

Title - Solving business problem for Veenus as a professional consultant

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What is the key problem of the company?

The challenges faced by Veenus include motivating partners for customer experience, establishing win-win scenarios with service partners, encouraging partners digital onboarding, and enhancing brand image on social media.

Challenges

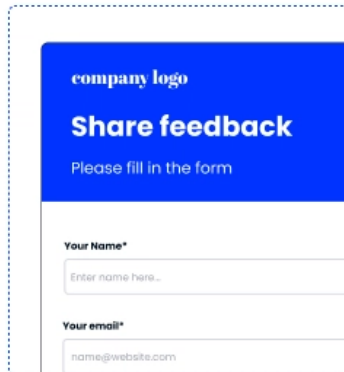
1. Encouraging service partners for exceptional customer experience while simultaneously establishing mutually beneficial scenarios that cater to both the partner's and the company's interests is the primary hurdle faced by Veenus.
2. Secondly, promoting booking partner's digital onboarding is another one of the challenges faced by Veenus. As stated by the owner of the company, Veenus is left with only 30% of partners post COVID-19 and they're finding it difficult to motivate digital onboarding.
3. Enhancing brand image on social media is the most important aspect Veenus is struggling with. As Veenus went digital, it is looking to focus on discoverability on digital platforms.

Proposed solution

Our first solution is to gather feedback and reviews from booking partners and to achieve this we suggested Veenus to create a separate dashboard for booking partners where they can view their past bookings and provide feedback for each experience. The dashboard can include a feedback widget to track experience which can include feedback fields like quality, reliability, communication, pricing, timeliness and service.

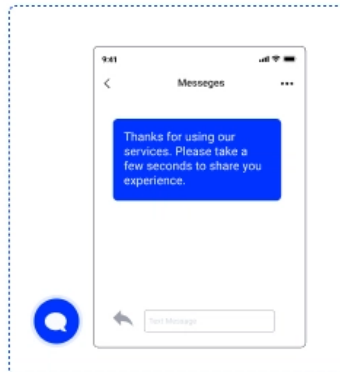
Tools for customer feedback

Survey



A sample survey form with a blue header containing a company logo and the text 'Share feedback' and 'Please fill in the form'. Below the header are two input fields: 'Your Name*' with a placeholder 'Enter name here...' and 'Your email*' with a placeholder 'name@website.com'.

Live chat



Feedback button

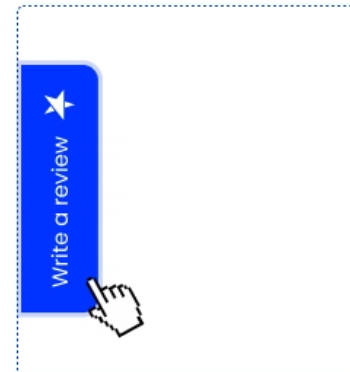


Fig 1: Sample survey form and feedback button

Once we have our setup for a feedback system, we can enrich partners' experience by suggesting them pre-curated packages. These packages will guarantee a good experience for customers and higher margins and good feedback for the company. We also suggested, they add positive messaging on their website which are crucial for building trust and confidence among new customers. Adding FAQ's and chatbot will also help them to reduce touchpoint and also make it convenient for potential customers to answer their common queries.

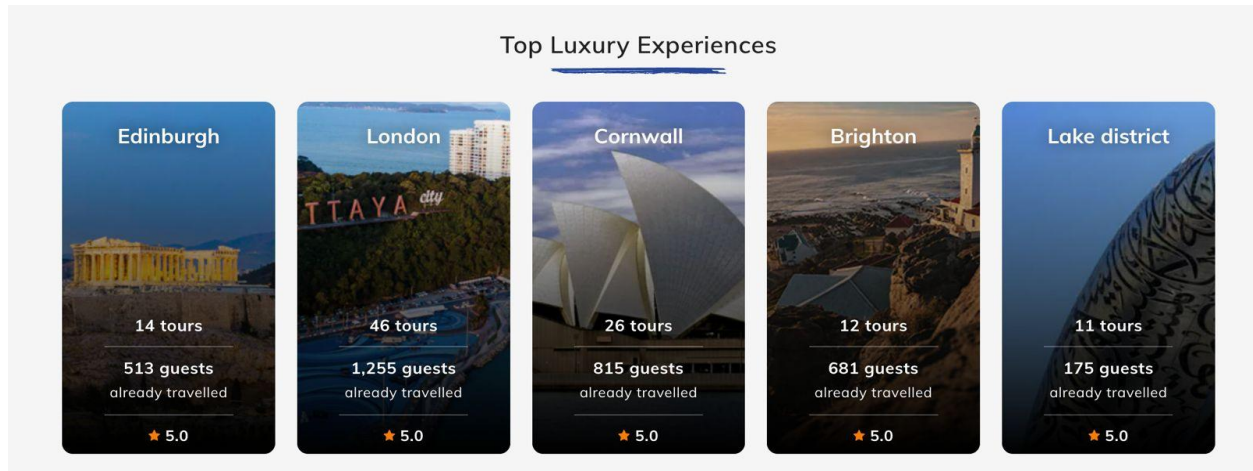


Fig 2: Sample pre-curated packages that have high margins and good service

Furthermore, to bridge service partners' experience, Veenus can offer incentive or rewards for outstanding performers. The company can also display names of existing service partnerships in a dedicated featured section for partners which will help to build trust and create win-win situations

Adding testimonies from existing service partners will also help the company to build trust among booking partners and build credibility for digital onboarding. By displaying business achievements on websites, Veenus can create FOMO(fear of missing out) and attract more partners.

In our research, we noticed that Veenus has very little online presence. By increasing online presence we can reach many people worldwide and break geographical barriers. Veenus can make sure to reach these kinds of audience by increasing the company's online presence and uploading engaging and informative content for travellers on their social media platform.

Why is your team's proposal useful for the company?

Main action points

Firstly, we can implement a Feedback and Rating system by creating a separate dashboard on their website for their booking partners. To conduct this plan, the company can follow agile practices which involve a continuous iterative approach that presents feasible output periodically. Specialised teams can work in squads that include few key members from each domain like business, product, design and engineering. These squads can work in a sprint fashion which usually has a duration of 2-3 weeks.

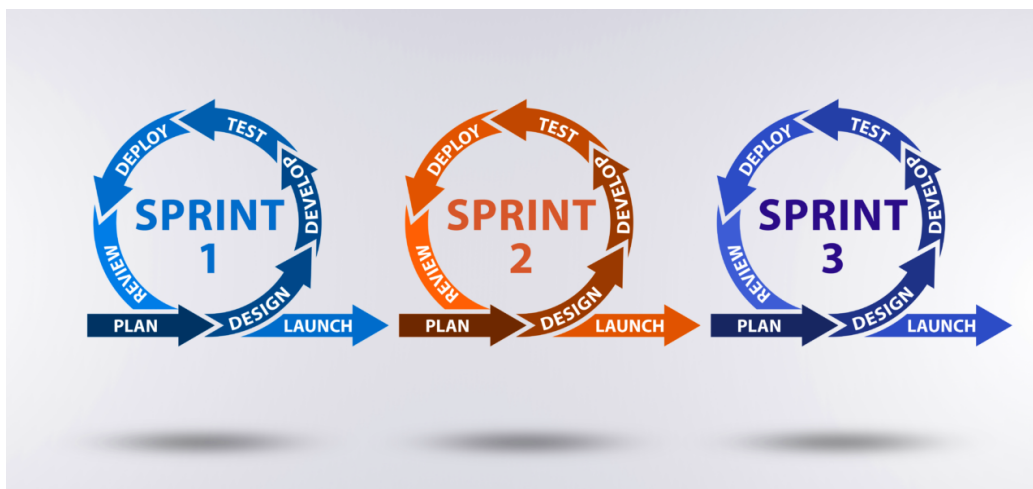


Fig 3: Lean Agile process with iterative launches and feedback

Teams can identify specific requirements and design for the features and functionality for the dashboard. The objectives and their success metrics can be discussed with stakeholders. And with continuous feedback and validation from stakeholders, the team can reach their objective.

Our second most important suggested action point is to improve online presence and social media. These days many companies are using tools to post content periodically that also provide real-time analytics and engagement for posts from various social media platforms (e.g Hootsuite,

Buffer.io). The company can make one-off investments that can be made with top influencers to collaborate based on their reach, engagement rate and content quality to drive more traffic to their website and content..

Benefits to the company

We have identified many weak points within the company and when conducting SWOT analysis which are, no feedback and review setup, not making the most of tools analytic tools that can track user behaviour and improve experience.

Veenus can now work on the suggestions, which will bring them at par with their competitors. Product improvements like a feedback system will help them gain insights to their user behaviour. On the business side they can create a bundle and customise experiences as the luxury travel business are high margin affairs. Social media engagement on the other hand will provide them the much needed discovery of their offerings.

Reflections on your work as a professional consultant

Approach of working with client

As Veenus is a small firm and they are not self aware about the root cause of the problem they are having, we would like to take the doctor-patient model of business consultancy. This model is like a close, collaborative diagnosis and treatment plan tailored specifically to the firm's unique needs and challenges. We'll co-create a personalised action plan and not just recommend any off-the-shelf solutions. The company can actively engage, asking questions and suggestions throughout the implementation of the solution. By closely working with Veenus, their team can absorb some learnings that we come across together as a team.

Key lessons and challenges while working as a professional consultant

The business problem with Veenus was unique and the learnings can be categorised under following key points:

Importance of Customer Feedback: Veenus used to operate on feedback which is received when something goes wrong, so they aren't aware of the good parts of their application. Implementing a feedback system from booking partners provides invaluable insights into customer preferences and service quality.

Product Innovation: Veenus has limited product/tour offerings on their website. These are atomic in nature and don't provide much details about the experience. Designing pre-curated packages based on feedback ensures that offerings are always aligned with customer desires and market trends.

Digital presence & Engagement: The company does not have any significant following, subscriptions on any social media platforms. An online presence acts as a trustable data point while making such decisions. And to reach a wider audience, the company needs to self promote their wins so that it encourages their service partners to compete and outperform each other.

What are your reflections about working as a team?

Experience while working as a team

Working in a team always carries its pros and cons. As a team we were able to brainstorm and come up with many ideas that could help Veenus identify the problems and work on the solutions. But as a professional consultant we also had to prioritise main action items so that we can provide high impact options as a solution to the company. We had multiple back and forth discussions to identify the key recommendations, we backed our ideas with data to ensure everyone is agreeing to the justification behind the approach.

Personal contributions to the project

We worked in a team of 4 members. The initial brainstorming sessions were a combined effort, after which we decided to pick one track to work on. I personally took part in product innovation where I identified areas of improvement needed in the website as it was a key component for this business. I suggested creating a central feedback system that would cater both the booking and

service partners. The product requirements involved adding a feedback interface to gather ratings and testimonials to build trust. A combination of a feedback system and travel bundles would be a business decision that we can make to design pre-curated packages. My personal experience as a digital marketing intern, helped me to connect the dots and provide solutions with minimal development changes for the engineering team using the tools that are freely available in the market.

Learnings from Q&A session

A couple of questions were asked after the presentation pertaining to:

- What is the use of webinar and who will it help ?

I believe we didn't clearly explain in the presentation that the webinars will be part of a regular process which we will set up with service partners so that we can answer their questions related to digital onboarding.

- How will we provide VR experience to users (specially since most booking partners may not own it)

This is again something which we incorrectly assumed. The cost of VR headsets is high and it is improbable that a booking partner would be having it at the time of booking. Looking back at it, I now have an alternative to capture and display 3D video which does not require any special device but provides an intermediate experience of reality.